

```
In[1]:= Clear["Global`*"];
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0. Readme

1. (x', y', z') are represented by $(x3, y3, z3)$.
2. $V_{\lambda\lambda\lambda}^S, V_{\lambda\lambda\phi}^S, V_{\lambda\lambda r}^S, V_{\phi\phi\lambda}^S, V_{\phi\phi\phi}^S, V_{\phi\phi r}^S, V_{rr\lambda}^S, V_{rr\phi}^S, V_{rrr}^S$ in Table 2 and Eqs. (20-22) are represented by SphericalV $\lambda\lambda\lambda$, SphericalV $\lambda\lambda\phi$, SphericalV $\lambda\lambda r$, SphericalV $\phi\phi\lambda$, SphericalV $\phi\phi\phi$, SphericalV $\phi\phi r$, SphericalV $rr\lambda$, SphericalV $rr\phi$, and SphericalV rrr .
3. The integral kernels are considered and the complete expressions should be presented as $G\rho \int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} \kappa(\text{Kernel}) d u' d \phi' d \lambda'$.

1. Basic settings

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In[2]:= A = x3 * Sin[λ] - y3 * Cos[λ];
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B = x3 * Cos[λ] + y3 * Sin[λ];
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In[4]:= l = Sqrt[r^2 + x3^2 + y3^2 + z3^2 - 2 * r * (z3 * Sin[φ] + B * Cos[φ])];
```

2. Expressions from Table 2 involved in Eqs. (20-22)

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In[5]:= SphericalVλλλ =
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$$3 * A * (6 * r^2 + x3^2 + y3^2 + 6 * z3^2 + 5 * (x3^2 - y3^2) * \cos[2 * \lambda] - 12 * r * z3 * \sin[\phi] - 12 * r * B * \cos[\phi] + 10 * x3 * y3 * \sin[2 * \lambda]) / (2 * l^7);$$

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In[6]:= SphericalVλλφ = -3 * (z3 * Cos[φ] - B * Sin[φ]) *
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$$(2 * r^2 - 3 * (x3^2 + y3^2) + 2 * z3^2 + 5 * (x3^2 - y3^2) * \cos[2 * \lambda] - 4 * r * z3 * \sin[\phi] - 4 * r * B * \cos[\phi] + 10 * x3 * y3 * \sin[2 * \lambda]) / (2 * l^7);$$

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In[7]:= SphericalVλλr =
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$$3 * (r * \cos[2 * \phi] * (x3^2 + y3^2 - 2 * z3^2 + (x3^2 - y3^2) * \cos[2 * \lambda] + 2 * x3 * y3 * \sin[2 * \lambda]) + B * \cos[\phi] * (-6 * r^2 + 3 * (x3^2 + y3^2) - 2 * z3^2 - 5 * (x3^2 - y3^2) * \cos[2 * \lambda] + 8 * r * z3 * \sin[\phi] - 10 * x3 * y3 * \sin[2 * \lambda]) + 2 * r * (r^2 - x3^2 - y3^2 + 2 * z3^2 + 3 * (x3^2 - y3^2) * \cos[2 * \lambda] + 6 * x3 * y3 * \sin[2 * \lambda]) - z3 * \sin[\phi] * (6 * r^2 - 3 * (x3^2 + y3^2) + 2 * z3^2 + 5 * (x3^2 - y3^2) * \cos[2 * \lambda] + 10 * x3 * y3 * \sin[2 * \lambda])) / (2 * l^7);$$

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In[8]:= SphericalVφφλ =
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$$3 * A * (r^2 + x3^2 + y3^2 + z3^2 - 5 * z3^2 * (\cos[\phi])^2 - 2 * r * z3 * \sin[\phi] - 2 * r * B * \cos[\phi] + 5 * z3 * B * \sin[2 * \phi] - 5 * B^2 * (\sin[\phi])^2) / l^7;$$

```
In[9]:= SphericalVφφφ =
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$$3 * (z3 * \cos[\phi] - B * \sin[\phi]) * (-3 * (r^2 + x3^2 + y3^2 + z3^2) + 5 * z3^2 * (\cos[\phi])^2 + 6 * r * z3 * \sin[\phi] + 6 * r * B * \cos[\phi] - 5 * z3 * B * \sin[2 * \phi] + 5 * B^2 * (\sin[\phi])^2) / l^7;$$

```

In[10]:= SphericalVφφr = 3 * (5 * z3^2 * B * (Cos[φ])^3 + 5 * z3 * B^2 * (Sin[φ])^3 -
      z3 * Sin[φ] * (3 * r^2 + x3^2 + y3^2 + z3^2 + 5 * y3 * z3 * Sin[2 * φ] * Sin[λ]) +
      r * (r^2 + x3^2 + y3^2 + z3^2 + 7 * z3 * B * Sin[2 * φ]) + r * (Sin[φ])^2 * (2 * z3^2 - 5 * B^2) +
      (Cos[φ])^2 * (5 * z3^2 * (-r + z3 * Sin[φ]) + 2 * B^2 * (r - 5 * z3 * Sin[φ])) -
      Cos[φ] * (-5 * x3^3 * (Sin[φ])^2 * (Cos[λ])^3 -
      15 * x3^2 * y3 * (Sin[φ])^2 * Sin[λ] * (Cos[λ])^2 + y3 * Sin[λ] *
      (3 * r^2 + x3^2 + y3^2 + z3^2 - 5 * y3^2 * (Sin[φ])^2 * (Sin[λ])^2) + x3 * Cos[λ] *
      (3 * r^2 + x3^2 + y3^2 + z3^2 + 5 * (Sin[φ])^2 * (2 * z3^2 - 3 * y3^2 * (Sin[λ])^2)))) / l^7;

In[11]:= SphericalVrrλ = -3 * A * (4 * r^2 - x3^2 - y3^2 - z3^2 +
      (z3 * Sin[φ] + B * Cos[φ]) * (-8 * r + 5 * z3 * Sin[φ] + 5 * B * Cos[φ])) / l^7;

In[12]:= SphericalVrrφ = -3 * (-z3 * Cos[φ] + B * Sin[φ]) * (4 * r^2 - x3^2 - y3^2 - z3^2 +
      (z3 * Sin[φ] + B * Cos[φ]) * (-8 * r + 5 * z3 * Sin[φ] + 5 * B * Cos[φ])) / l^7;

In[13]:= SphericalVrrr =
      -3 * (5 * (r - z3 * Sin[φ] - B * Cos[φ])^3 - 3 * l^2 * (r - z3 * Sin[φ] - B * Cos[φ])) / l^7;

```

3. Test the consistency of Eqs. (20-22)

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In[14]:= Laplace1 = FullSimplify[SphericalVλλλ + SphericalVφφλ + SphericalVrrλ]
Out[14]=
0

In[15]:= Laplace2 = FullSimplify[SphericalVλλφ + SphericalVφφφ + SphericalVrrφ]
Out[15]=
0

In[16]:= Laplace3 = FullSimplify[SphericalVλλr + SphericalVφφr + SphericalVrrr]
Out[16]=
0

In[17]:= NotebookSave[EvaluationNotebook[]];

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